Land Cover Classification based on NDVI using LANDSAT8 Time Series: A Case Study Tirupati Region

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***Abstract*—The Normalized Difference Vegetation Index (NDVI) is one of the most widely used numerical indicator that uses the visible (VIS) and near-infrared bands (NIR) of the electromagnetic spectrum, and is utilized to analyze remote sensing images and assess whether the target contains live green vegetation or not. This paper analyses the utility of NDVI for mapping the land cover characteristics over Tirupati Region, Chittoor District, Andhra Pradesh, India. Images of level-1 data of Landsat8 were collected for three different seasons of the same region and derived the NDVI values. For land cover classification, various band combinations of the remote sensed data are treated and the spatial distribution such as water bodies, built-up area, vegetation and dense vegetation are easily examined by computing their normalized difference vegetation index from multi-spectral data. On-going portion of present study focuses on making out the difference between the vegetation indexes of different land cover types by performing supervised classification.**

***Index Terms*—Remote Sensing, Image Classification, Multispectral Image, Normalized Difference Vegetation Index (NDVI).**

I. INTRODUCTION

Land cover mapping is one of the major important applications of remote sensing data. Land covers are the regular contributors of the physical condition of the ground surface. Satellite based remote sensing has helped in study of land cover dynamics. The number of remote sensing platforms increased during the past decades and produced more detailed geographical data [1].

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Sensors can supply accurate and timely information on land cover, its location and spatial configuration, as well as the growth rates. Land cover mapping from remote sensing data is based on image classification. Recent advancement in the sensor capabilities having better spatial and spectral resolution is a major breakthrough in satellite remote sensing. Both visual and digital analysis techniques are used for mapping and monitoring forest/vegetation. However, it is necessary to select proper methodology for interpreting various land use classes such as forest, vegetation classes [2].

Three factors distinguish the analysis of remotely sensed imagery through interpretation. They are: (1) Remotely sensed images usually provide a panoramic overview, (2) Remote sensing images use the visible and infrared region of the electro-magnetic spectrum and (3) Remote sensing images can portray the Earth's surface at different scales and resolutions.

Multispectral remote sensing images contain the necessary spectral and spatial features of the various objects [3]. Classification of the objects is done through spectral analysis of the radiant energy reflected or emitted by the target. In this paper, multispectral images are used for finding the Normalized Difference Vegetation Index (NDVI) values and classifying different land cover types over the selected study area. The NDVI is based on the difference between the at-sensor spectral radiance of the red band (band4) and the near-infrared band (band5) of satellite image. Theoretically the values of the NDVI vary between –1.0 and +1.0, but are commonly positive for soil and vegetation [4].

To present the stated objective this paper is outlined into 5 sections, where section II present the Description of Study Area and Data, Methodology is presented in section III. Section IV outlines the obtained experimental results for the proposing approach. The concluding remarks are outlined in section V

II. DESCRIPTION OF STUDY AREA AND DATA

Tirupati is a most visited pilgrim city in Chittoor district of the Indian state of Andhra Pradesh. It is the largest urban cluster in the Rayalaseema region. It is geographically situated on latitude 13° 35′ 19″ North to 13° 42′ 10″ North and on longitude 79° 31′ 6″ East to 79° 35′ 5″ East as per the datum – WGS84.The city is surrounded by seven hills and is

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situated at the foot of Eastern Ghats which were developed during Precambrian era. One of its outlying cities, Tirumala which is famous for Sri Venkateswara Temple is located within these hills.

Geologically the Tirupati - Tirumala Ghat road at 12 km point shows a lack of continuity of stratigraphic significance that represents a period of remarkable tranquility. This is cited as Eparchaean Unconformity [Source: Tirupati Urban Development Authority]. In Tirupati, monsoon remains moderate and in summer, city experiences temperatures ranging from 35 to 41 degrees Celsius and in winter the minimum temperatures will be around 16 to 20 degrees Celsius. The summer season is from March to June, with the arrival of rainy season in July to December followed by winter till the end of February. The study area experiences maximum rainfall in October to November during northeast monsoon season [Source: Regional Agricultural Research Station].

The multispectral remote sensing images of Tirupati region of three different seasons were collected from USGS (United States Geological Survey). Landsat 8 satellite images the entire earth once every 16 days. The sensors called OLI (Operational Line Imager) and TIRS (Thermal Infrared Sensor) provide data in multispectral bands. Band designations of Landsat8 are as given in Table I. Satellite data over Tirupati region of 21st January, 30th May and 21st October of 2015 (day time, level-1G product, path/row 143/51) have been used in this study.

TABLE I
LANDSAT8 OLI AND TIRS

Band Designations	**Wavelength (µm)**	**Resolution (m)**
Band 1 (Coastal Aerosal)	0.43 - 0.45	30
Band 2 (Blue)	0.45 - 0.51	30
Band 3 (Green)	0.53 - 0.59	30
Band 4 (Red)	0.64 - 0.67	30
Band 5 (Infrared)	0.85 - 0.88	30
Band 6 (Short wave infrared)	1.57 - 1.65	30
Band 7 (Short wave infrared)	2.11 - 2.29	30
Band 8 (Panchromatic)	0.50 - 0.68	15
Band 9 (Cirrus)	1.36 - 1.39	30
Band 10 (Thermal infrared)	10.6 - 11.2	100
Band 11 (Thermal infrared)	11.50 - 12.51	100

III. METHODOLOGY

The satellite data products were geometrically corrected data set. The DN (Digital Numbers) values were converted to at-sensor spectral radiance of each band for Landsat8 [5] using

$$L_\lambda = \frac{(L_{max} - L_{min}) * Q_{cal}}{(Q_{calmax} - Q_{calmin})} + L_{min} \qquad (1)$$

Where
L_{max} is the maximum at-sensor spectral radiance ($Wm^{-2}sr^{-1}\mu m^{-1}$)
L_{min} is the minimum at-sensor spectral radiance ($Wm^{-2}sr^{-1}\mu m^{-1}$)
Q_{cal} is the DN value of pixel
Q_{calmax} is the maximum DN value of pixels
Q_{calmin} is the minimum DN value of pixels

The Visible bands and Near Infrared bands are combined together to form an FCC (False Color Composite) image. The images were resampled using nearest neighbor method. All the data are re-projected to a Universal Transverse Mercator (UTM) coordinate system, datum WGS84, zone 44. NDVI can be calculated from visible and infrared bands of Landsat8 spectral bands. NDVI is calculated on per-pixel basis as the normalized difference between the red band (0.64 - 0.67µm) and near infrared band (0.85-0.88µm) of the images using the formula [4].

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)} \qquad (2)$$

Where NIR is the near infrared band value of a pixel and RED is the red band value of the same pixel.

Training Sets were collected from FCC imagery and supervised image classification using Maximum Likelihood Classification is carried for each dataset [6]. The statistics of NDVI values like minimum, maximum, mean and standard deviation for various land cover types of different seasons are compared. NDVI time series provide seasonal changes and same or similar curves of NDVI time series belong to the same category of land use/cover [7]. Time-series data of normalized difference vegetation index (NDVI) can be used to support land-cover change detection and phenological parameter derivations [8].

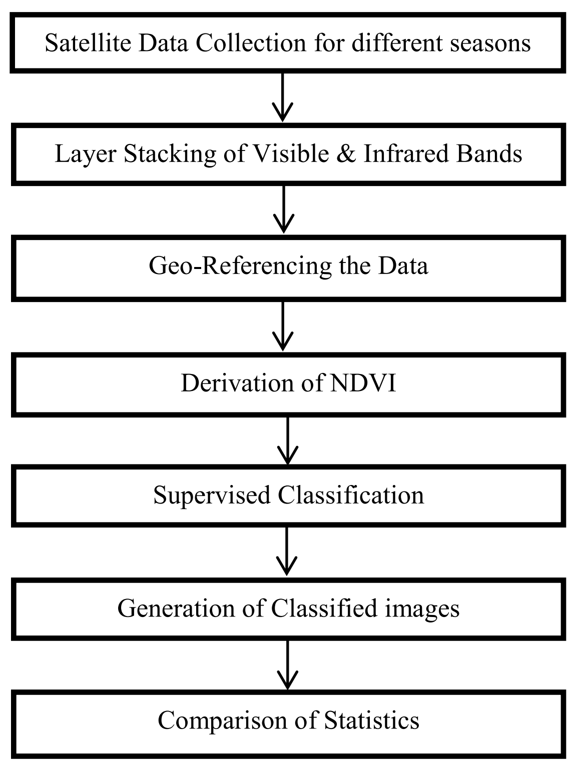

Fig. 1. Flow chart of the methodology

The approach to the proposed work to classify the image based on NDVI values is shown in the Figure1 and the statistics of NDVI are estimated.

IV. RESULTS & ANALYSIS

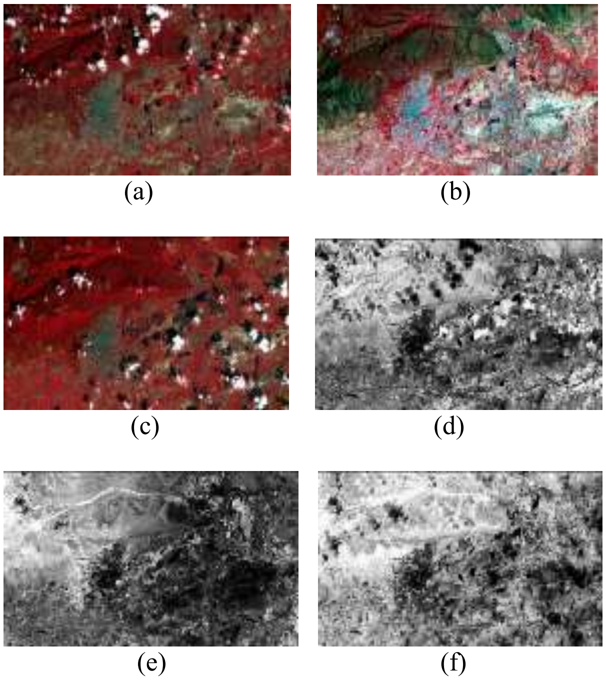

Fig. 2 (a), (b), (c) - False Color Composite of Satellite Imagery acquired on 22nd January, 30th May and 21st October; (d), (e), (f) – NDVI Image extracted for 22nd January, 30th May and 21st October

TABLE II
STATISTICS OF NDVI VALUES FOR DIFFERENT LAND COVER TYPES FOR 22.01.2015

Statistics	Minimum	Maximum	Mean
Built Up	-0.174	0.266	-0.024
Water Bodies	-0.496	-0.062	-0.328
Dense Vegetation	0.346	0.647	0.570
Vegetation	0.009	0.563	0.244
Bare Soil	-0.056	0.345	0.078

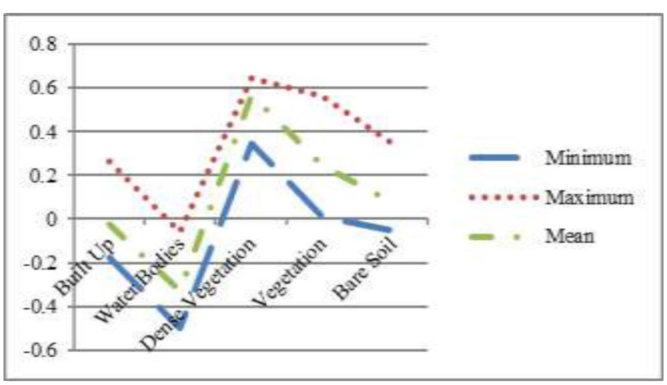

Fig. 3. Plot showing statistics of NDVI for 22.01.2015

TABLE III
STATISTICS OF NDVI VALUES FOR DIFFERENT LAND COVER TYPES FOR 30.05.2015

Statistics	Minimum	Maximum	Mean
Built Up	-0.172	0.079	-0.060
Water Bodies	-0.182	-0.054	-0.145
Dense Vegetation	0.293	0.615	0.500
Vegetation	0.151	0.521	0.319
Bare Soil	-0.082	0.138	-0.001

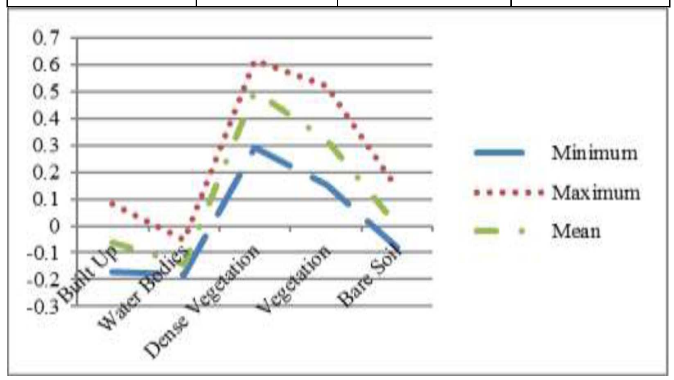

Fig. 4. Plot showing statistics of NDVI for 22.01.2015

TABLE IV
STATISTICS OF NDVI FOR DIFFERENT LAND COVER TYPES FOR 21.10.2015

Category	Minimum	Maximum	Mean
Built Up	-0.273	0.287	-0.019
Water Bodies	-0.332	-0.011	-0.175
Dense Vegetation	0.445	0.657	0.575
Vegetation	0.00	0.629	0.404
Bare Soil	0.006	0.467	0.166

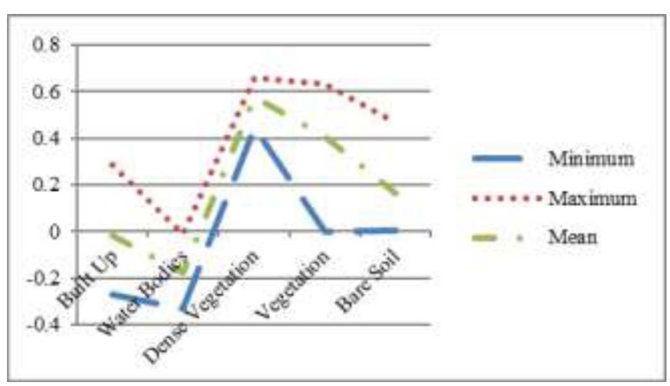

Fig. 5. Plot showing statistics of NDVI for 22.01.2015

TABLE V
NDVI VALUES OF DIFFERENT LAND COVER TYPES FOR THE THREE DATES.

Land Cover Types	22.01.2015	30.05.2015	21.10.2015
Built Up	-0.024	-0.060	-0.019
Water Bodies	-0.328	-0.145	-0.0175
Dense Vegetation	0.570	0.500	0.575
Vegetation	0.244	0.319	0.44
Bare Soil	0.078	-0.001	0.166

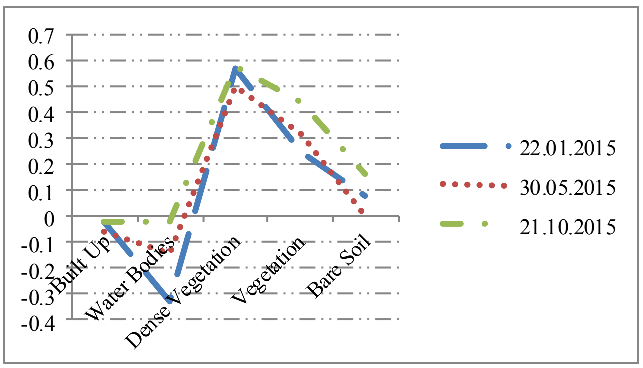

Fig. 6. Plot showing NDVI values of Different Land Cover types for the three dates.

The statistics show that lesser the NDVI value usually represent water bodies (ranging from -0.0175 to -0.328), Built up (ranging from -0.019 to 0.060) and bare soil (ranging from -0.001 to 0.166). The NDVI value of the dense vegetation is ranging from 0.500 to 0.575 which is almost consistent throughout the three seasons. This indicates forest region of Eastern Ghats. The vegetation in and around the city varies between 0.244 and 0.44 for the three seasons.

V. CONCLUSION & FUTURE SCOPE

In this paper, multi season and multispectral satellite data has been made use of to classify the study area on the basis of NDVI. It is observed that the Eastern Ghats, spread over Tirupati region has dense vegetation in the forest range. The eastern part of Tirupati has plenty of water bodies. Both Eastern and Western parts have vegetated lands. Most of the central part is covered with built up area and settlements. The three seasons study indicates the change in NDVI value of vegetation and its dynamics, even though there is no much change in dense vegetation.

This methodology can be extended to retrieve the non-spatial parameters that can affect the vegetation through climatic changes over the study area. NDVI can be used to estimate the Land Surface Emissivity (LSE) and Land Surface Temperature (LST) and can be derived the correlation between NDVI and LST.

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